

NCO

Goal

Prepare IP for generate 2048 samples of waveform (sinus, triangle, or rectangle).

Provided module

You can edit only the *task_13.sv* file. If you wish, you may add other files for supportive submodules.

The module has the following inputs and outputs:

Table 1 List of inputs and outputs of the task_13 module.

Signal name	Signal type	Bit length
i_clk	Input	1 bit
i_rst	Input	1 bit
i_valid	Input	1 bit
i_first	Input	1 bit
i_data	Input	16 bits
i_last	Input	1 bit
o_valid	Output	1 bit
o_data	Output	16 bits
o_last	Output	1 bit

The reset is synchronous and active high.

Input and output interfaces

In one packet task receives two configuration words. First received word is a number of periods generated wave, second byte is the shape of the wave.

Task receives data in the way shown in the figure below:

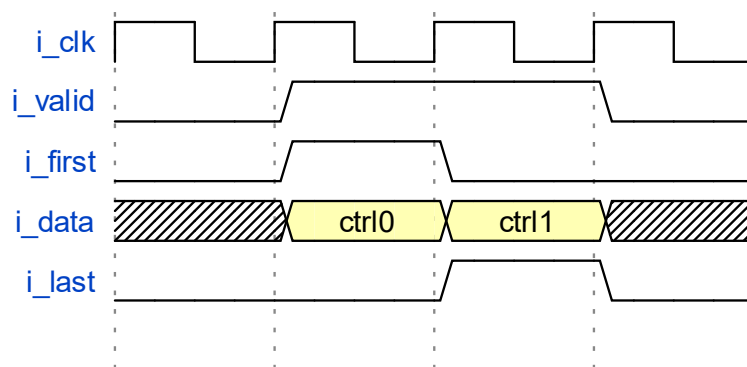


Figure 1 Input waveform .

ctrl0 – number of periods, N is a **power of two** between 1 and 512

ctrl1 – shape coding: 01 – sinus
 02 – triangle
 03 – rectangle

Value of the wave (amplitude) generated by the IP is between -32767 and 32767 for sinus, and between 0 and 32767 for triangle and rectangle.

To send data from the task in a proper way one needs to provide the *o_valid* signal that is synchronized with data on the *o_data* bus and *o_last* signal that is synchronized with the last sample in output packet. The IP should be responded by sending a wave samples (always 2048 samples).

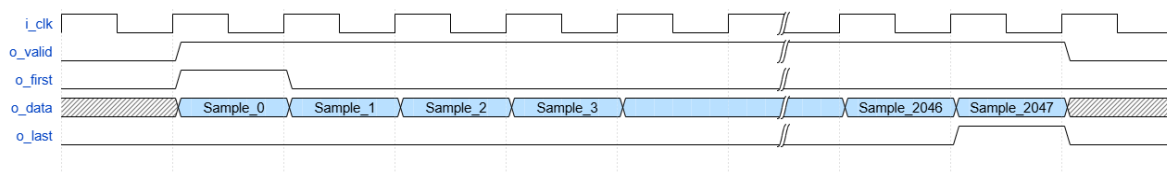


Figure 2 Output waveform

Output data can be sent as a continuous stream, or it can have regular or irregular gaps. The only requirement is providing *o_valid* and *o_last* signals.

Examples

Example 1: – generate 2 period of sinus

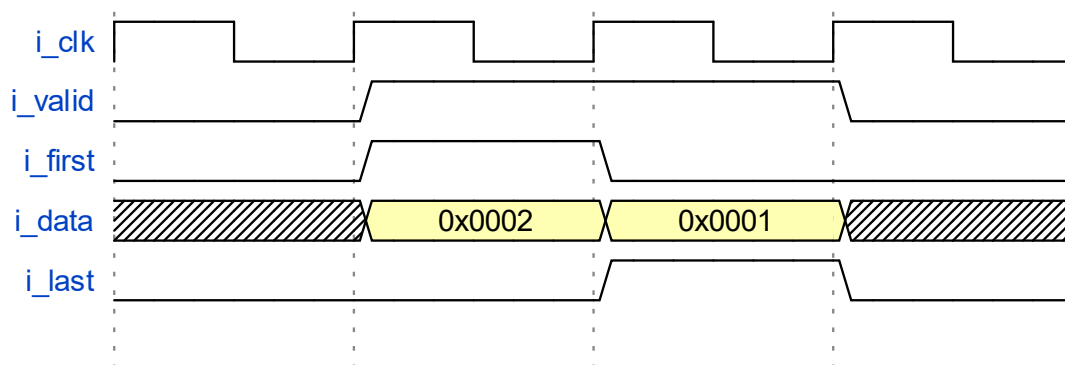


Figure 3 Input packet

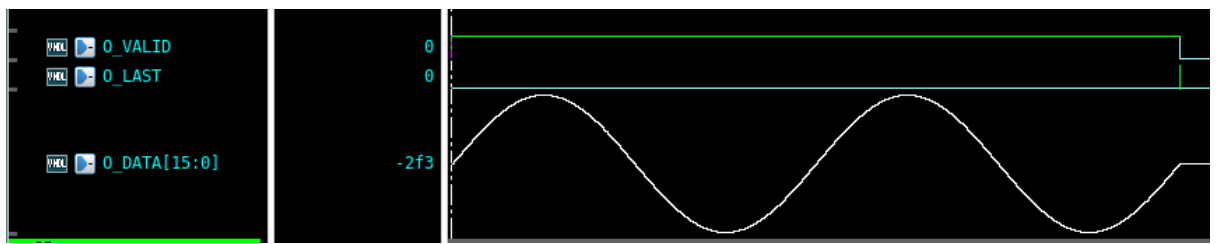


Figure 4 Output packet

Example 2: 3 periods of triangle

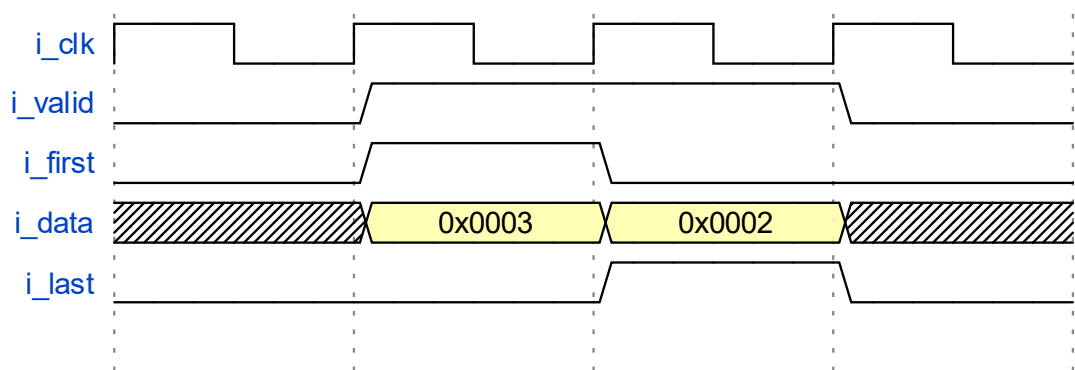


Figure 2 Input packet

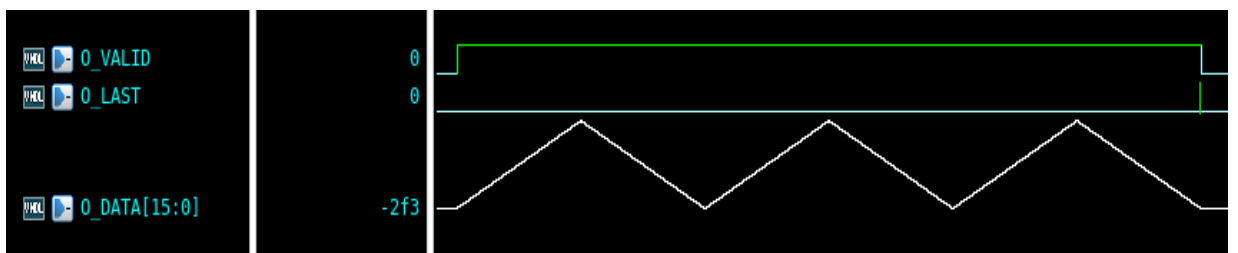


Figure 4 Output packet

Example 3: 2 periods of rectangle

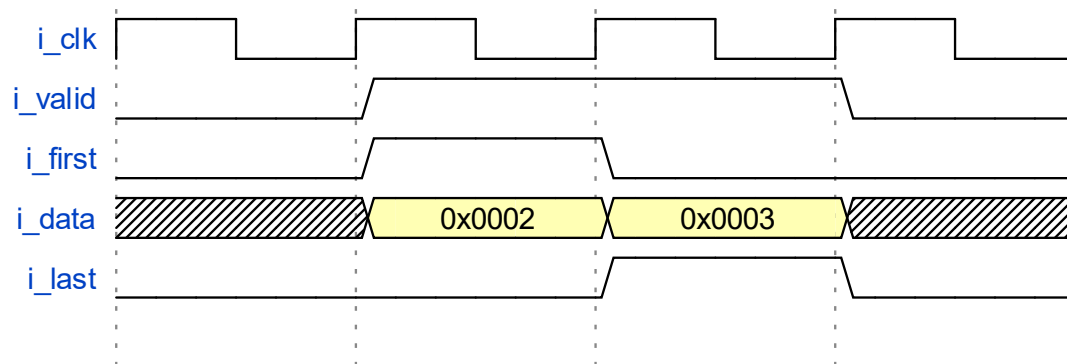


Figure 2 Input packet



Figure 2 Outout packet

About Development mode and Evaluation mode check chapter 2.3.1 **Testing modes** in the **FPGA_Hackathon_2026_Project_Guide** document.

Task will be tested with 3 gold vectors and pointed 4 points per vector.

Additional 5 points may be awarded for resource utilization if latency does not exceed 21μs (for more information, check chapter 2.3.2 **Bonus Points** in the **FPGA_Hackathon_2026_Project_Guide** document).